

Lab 2: KNN and Naïve Bayes Classifiers

Evaluation: 5% of the total grade

In this lab exercise you will work with the breast-cancer dataset from sklearn. There are 30 diagnostic measurements (descriptive features) and two classes in the target feature (0: “malignant”, 1: “benign”) in this dataset. The objective is to predict whether or not a patient has the breast cancer based on the diagnostic measurements. Details of the dataset can be found [here](#).

Your task is to train and test classification models with different algorithms (KNN & Naïve Bayes) and compare performance on the test dataset using multiple measures of performance as discussed in the class.

You will also explore if a smaller set of selected features can be used as input to a model without compromising the model performance. You will develop classification models first using all the features and then using only the selected features (by using SequentialFeatureSelector, for example).

Perform the following tasks:

1. Load and explore the dataset for data quality issues
2. Prepare the data for model development. Examining the feature correlations maybe a good starting point.
3. Train and evaluate KNN and Naïve Bayes classifiers to predict whether or not the patients have the disease
4. Report and briefly discuss the performance of the model on an independent test set based on the following measures of performance:
 - a. Classification report
 - b. Confusion matrix
5. Briefly discuss the performance of each model on the test set.
6. Add a brief summary comparing the performance of these models in terms of accuracy measures and execution times for model training and testing.

Submit your lab report along with the source code preferably as Jupyter notebook in pdf format by June 13, 2025 @ 11:59.

Lab Grading Criteria

Fraction	0.8 – 1	0.4 – 0.7	0 – 0.3	Max Score
Category				
Model development workflow	Followed all steps in the modeling process	Followed most steps in the modeling process	No order in the modeling process	1
Tasks and observations	Performed all the tasks correctly, results properly presented and well explained	Performed most of the tasks correctly, results fairly presented and explained	Performed some of the tasks correctly, poorly presented results with little explanation	3
Code	Well organized, easy to follow and clean code with meaningful comments	Somewhat organized, a little hard to follow and fairly clean code with some comments	Not organized, very hard to follow and messy code with little or no comments	1
Total				5